FoodWise AI

AI-Powered Food Waste Prediction, Surplus Detection & Redistribution Support Platform

IBM SkillsBuild Internship | SDG 12 — Responsible Consumption & Production | Target 12.3

PROBLEM	SOLUTION	UNIQUE VALUE PROPOSITION	UNFAIR ADVANTAGE	CUSTOMER SEGMENTS
• Food-serving orgs struggle to match preparation with fluctuating demand • Potential surplus often identified too late for effective action • Staff lack a structured, data-driven workflow for surplus decisions • Manual estimation leads to recurring over-preparation	• ML-based demand prediction (Random Forest Regressor) • Transparent Low/Medium/High surplus-risk scoring • Five-action recommendation engine (Monitor, Reduce Prep, Discount, Redistribute, Discard) • Simulated redistribution recipient matching • Sustainability insights dashboard	**Predict food demand, detect surplus early, and turn potential food waste into actionable decisions.** A lightweight, explainable AI platform that helps food-serving organisations act on surplus risk before food is wasted.	• Integrated end-to-end workflow: prediction → risk → recommendation → redistribution • Transparent, explainable risk logic — every decision can be explained • Lightweight Python/Streamlit stack — deployable without cloud infrastructure • Built for food-serving orgs, not generic analytics	• Restaurants & cafeterias • University/college canteens • Hotel food & beverage depts. • Institutional kitchens • Supermarkets (perishables) • Food-service organisations
EXISTING ALTERNATIVES	**KEY METRICS**	**HIGH-LEVEL CONCEPT**	**CHANNELS**	**EARLY ADOPTERS**
• Manual demand estimation • Spreadsheets / basic POS reports • Traditional inventory software • Informal surplus coordination • No systematic surplus workflow	• Demand prediction: MAE, RMSE, R^2 • Predicted surplus quantity per item • Waste-risk distribution (L/M/H) • Redistribution-eligible surplus units • No. of surplus alerts / recommendations *Prototype results on synthetic/demo data: MAE 5.80 | RMSE 7.49 | R^2 0.797 QA audit: 98 PASS, 0 FAIL (100 checks)*	An AI-powered decision-support platform combining demand forecasting, surplus-risk analysis, action recommendations and redistribution support — for food-serving organisations. *Like a "weather forecast for food waste" — identifying risk early enough to act.*	• Web app (Streamlit, browser-based) • Direct institutional partnerships • Pilot programmes with canteens/hotels • University sustainability programmes • Demonstration workshops	• University cafeterias (high daily volume) • Restaurants with significant preparation • Institutional kitchens & catering • Hotels with buffet/catering operations • Orgs with sustainability mandates
COST STRUCTURE		**REVENUE STREAMS**		
• Software development & maintenance • Cloud hosting & data storage • ML model development & retraining • API integrations (future) • Deployment & customer support		• SaaS subscription tiers (future) • Enterprise licences (future) • Institutional analytics plans (future) • Custom deployment & support (future) *Revenue streams represent potential future business models, not current project income.*		

Decision-support prototype only. All metrics are prototype results on synthetic/demo data. Redistribution data is simulated. Food safety is not autonomously certified by this system.